

Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.	(a)			canary yellow / bright yellow	1			1		1
	(b)			$\text{Pb}^{2+}(\text{aq}) + 2\text{I}^{-}(\text{aq}) \rightarrow \text{PbI}_2(\text{s})$ STATE SYMBOLS MUST BE SHOWN	1			1		1
2.				iron and Haber process / making ammonia / appropriate equation nickel in hydrogenation of alkenes	1			1		
3.				it is soluble because the enthalpy change of hydration is more exothermic (or more negative) than the enthalpy change of lattice making OR calculation of enthalpy of solution as -141 kJ mol^{-1} (1) the enthalpy of solution would be negative / exothermic (1)		1 1		 2		
4.				award (1) for every two correct - platinum electrode $1 \text{ mol dm}^{-3} \text{ H}^{+}(\text{aq})$ or any appropriate strong monobasic acid $\text{H}_2(\text{g})$ at 1 atm / $1.01 \times 10^5 \text{ Pa}$ - temperature = 298 K / 25°C (units must be included)	2			2		2
5.	(a)			35 and 17.5		1		1	1	1
	(b)			rate at 0.200 = 0.0286 rate at 0.400 = 0.057 ecf from part (a)		1		1	1	1
6.				copper metal, water, nitrogen gas, air		1		1		
Section A total					5	5	0	10	2	6